

1. Personal vs. Professional Data

- **Personal:** Photos, contacts, messages, health records.
- **Enterprise:** Corporate email, VPN profiles, proprietary documents.

2. Static vs. Dynamic Data

- **Static:** Photos, videos (infrequently updated).
- **Dynamic:** App settings, notes, SMS logs (frequently changing).

3. Criticality Levels

- **Tier 1:** Irreplaceable (contacts, two-factor secrets).
- **Tier 2:** Important but reproducible (documents, notes).
- **Tier 3:** Low value (cached app data, temp files).

Step-by-Step Classification Workshop:

1. Create a spreadsheet listing all data types per device.
2. Assign each row a tier and classify backup frequency accordingly.
3. Highlight any data requiring special compliance (e.g., GDPR).

Local backup strategies

Local backups provide the fastest recovery path when devices or networks fail.



iOS Local backups via Finder/iTunes

1. Connect and launch

- **Windows:** Open iTunes (or Apple Devices app) after plugging in your iPhone via USB-C/Lightning.
- **macOS:** Open Finder and select your device under Locations.

2. Enable encryption

- Check “Encrypt local backup,” then set and verify a strong backup password.

3. Initiate backup

- Click Back Up Now. Monitor progress; encrypted backups include Keychain, Health, and HomeKit data.

4. Archive critical snapshots

- In Finder, click Manage Backups, right-click the latest backup, and choose Archive to prevent auto-deletion.
Tip: Store archived `.backupdb` folders on an external SSD or network share for redundancy.

Android local backups via ADB and third-party tools

1. ADB Backup

- Enable Developer Options → USB Debugging, connect via USB, then run:
`adb backup -all -f mobile_backup.ab`
Use `android-backup-extractor` to convert `.ab` to tar.

2. Third-Party Tools

- **Titanium Backup (root):** Batch backup apps and data into `/sdcard/TitaniumBackup/`.
- **FolderSync:** Schedule nightly syncs of media and documents to a local NAS.

3. Secure Storage

- After backup, offload `.ab` or tar files to encrypted external storage (e.g., VeraCrypt container).

Laptop backups with external drives

- **Windows:** Use File History (Settings → Update & Security → Backup) to an external HDD, enabling versioning.
- **macOS:** Turn on Time Machine (System Settings → General → Time Machine) and select an external drive; ensure Encrypt backups are checked.
- **Cross-Platform Tip:** Use exFAT-formatted SSDs for compatibility; store the drive in a fire-safe or secure location.

Cloud backup solutions

Off-device backups guard against physical damage or theft. Key offerings include iCloud, Google One, and third-party SaaS.

7.3.1 iCloud backup (iOS / macOS)

1. Enable iCloud backup

- **iOS:** Settings → [Your Name] → iCloud → iCloud Backup → Back Up Now.

2. Storage management

- **Base free tier:** 5 GB; upgrade to iCloud+ for 50–2 TB plans.

3. Selective sync

- In Settings → [Your Name] → iCloud, toggle off non-critical apps to conserve quota.

Note: iCloud backups are end-to-end encrypted when Advanced Data Protection is enabled, covering all categories